Transfer: thermal energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose an energy-transfer model and communicate its limits.

SUCCESS CRITERIA - I can...

1. Distinguish warming, phase change and work.
2. Use correct signs and units.
3. Explain why a model result depends on its assumptions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A heat engine receives 500 J heat and outputs 150 J work per cycle. Find efficiency and rejected heat.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q = mc\,\Delta T$; $Q = mL$; $\Delta U = Q - W_{by}$; $\eta = W_{out}/Q_{hot}$. Select the applicable model and supplied constants. State heat-loss, phase-change and cycle assumptions.

Worked reasoning: A system takes 120 J of heat, does 50 J of work and increases internal energy by 70 J. An energy model must include transfers and storage rather than treating all heat as useful work.

Your turn: Draw its system boundary, input, output and storage change.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

A hot block and cool water are isolated together. Draw initial heat flow and describe equilibrium without inventing a final temperature.



Hot block



Cool water

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

Compare two heating simulations. Sample A has mass 2 kg and c = 100 J/(kg·°C); it warms by 4°C. Sample B has mass 2 kg and latent heat L = 300 J/kg; it completes a phase change at constant temperature. Label each process and find its energy input.

Sample	Process	Energy equation	Energy (J)
A	Warming	____	____
B	Phase change	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

An engine absorbs 500 J and produces 150 J work in a cycle. Find rejected heat and explain why an energy balance alone does not establish that any temperature pair can achieve that efficiency.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A device transfers 24 J in 4 s, then performs the same transfer in 8 s. Find both average powers and explain the difference.
Distinguish transferred energy from its transfer rate.

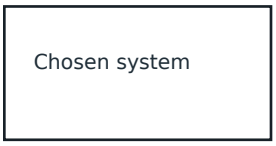
Case	Energy (J)	Time (s)	Power (W)
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

07 Exit check

A system absorbs 40 J and does 15 J work. Find ΔU using the unit convention. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



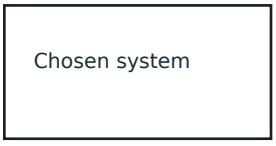
Stored energy change: _____ J

Marzano check - circle: ① ② ③

08 STRETCH: test the model

Design a second process with the same 70 J stored increase but a different work output. Explain why internal-energy change alone cannot determine heat input. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③